# Pricing rides as option contracts: guarantees and memberships under travel-time uncertainty

Mohamad Elmasri*

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*Department of Mathematics and Statistics, Carleton University`mohamadelmasri@cunet.carleton.ca`.

**Abstract**

Modern ride-share platforms must commit to a price before a trip is taken, yet the realized fare depends on travel time that is uncertain at the moment of sale, which can occur days in advance. The upfront price thus decomposes into the expected fare and the premium on an insurance claim whose payoff is the shortfall between the realized and promised prices. This work computes the fair premium for such a claim, following from contingent-claim pricing. Recent advances in travel-time distribution modeling cast the asymptotic distribution of travel time as a Brownian motion on a metric graph carrying a cyclostationary travel-time process. We thus derive closed-form, Black–Scholes-type premiums under both a population and a route-conditional travel-time law. A single pricing equation then covers products of increasing informational difficulty, such as on-demand and scheduled rides and capped-membership and commuter products. To price on sparse data, we develop route and travel-time samplers and validate them out of sample on 2014 Quebec City GPS trips, where they reproduce the empirical trip-time distribution. We show that guarantee premiums are inexpensive, typically under 5% of the fare. Conditioning on the realized route roughly halves the premium while reducing tail risk. Analyzing membership abuse, we show that adverse selection acts precisely along the dimension the price fails to condition on, yielding a design principle: subsidies additive in the fare, such as a fixed per-ride discount, leave marginal incentives unchanged and are abuse-proof by construction, with deterministic closed-form cost.

# 1 Introduction

## 1.1 Background

Mobility is vital to human activities, forming an integral component of our economic and trade networks, social interactions, political ties, and overall quality of life. Large-scale, trip-level data with extensive temporal and spatial coverage enables us to better diagnose transportation problems and develop effective solutions. Classical taxi providers priced rides based on actual time and distance traveled, with payment upon arrival. In contrast, modern ride-sharing services[1] must price rides well in advance. Commuter rides, for example, are priced at least a month in advance. Pricing more complex products, such as the right to travel a route for a predetermined fixed price, remains a challenge.

Two immediate challenges exist to achieve a general approach to pricing rides in transportation networks: understanding price variability in the marketplace and developing a modular, easily computable framework for pricing complex products. Price variability is predominantly influenced by travel uncertainty and supply and demand dynamics, the latter often controlled by ensuring sufficient driver availability. Recent progress in understanding travel uncertainty has demonstrated, both theoretically and in practical applications, that long-term deviations in travel time behave similarly to deviations from a normal distribution, despite the apparent chaotic nature of real-world traffic Elmasri et al. [2023], Woodard et al. [2017].

This work redefines the ride-share pricing problem as the pricing of a contingent claim, the right to a future trip at a promised price, whose value we compute with well-established financial tools. Building on the work of Elmasri et al. [2023], we price these agreements using

[1] Such as Waymo Inc., Lyft Inc., and Uber Inc.

well-established financial tools, providing a novel method to price complex transportation products, including, but not limited to, those mentioned earlier. We focus on strategically designing transportation incentives that reshape the commuter experience by offering financial guarantees that reduce pricing uncertainty[2]. In particular, we are interested in giving a definite answer to the following question:

> How to price a commuter membership that ensures a maximum trip cost of $K$?

## 1.2 Motivating example

We provide some intuition on the distribution of pricing that is partially observed in rides-share market using real-data. Consider the assumption a ride request is always fulfilled on route $\rho$ (sufficient supply of liquidity). Assume that rides do not interfere with each other. Let $P_\rho(t)$ be of such ride at start time $t > 0$. Varying the start time constitutes, hypothetically, a continuous stream of rides across $\rho$, and henceforth a realization of a continuous pricing function. Repeating this many times results in a continuous distribution of price over $\rho$.

Figure 1 illustrate this price action across two popular location in a given week. The offered prices in gray, and converted rides (actual paid) are in orange. A price distribution is apparent given the offer distribution. Inferring this distribution becomes harder the less popular the travel locations. Hence, to determine the upfront premium for capping the price of a ride at $K$ over a period $[0, T]$, we need to consider the dynamics of offers and and intent to request conversion which are not always available.

We provide some intuition on the price distribution that is only partially observed in a ride-share market, using real data. For a fixed route $\rho$, each departure time $t$ carries an offered price $P_\rho(t)$; sweeping $t$ over a period traces out a distribution of prices rather than a single value. Figure 1 shows this price action over one week. Offered prices are in grey, and the subset that converted to a paid ride is in orange. The grey spread makes the underlying price distribution visible, while the orange points show that we observe realized prices only where intent converted to a request, a partial selection-shaped view of the distribution. Inference becomes harder the less popular the origin–destination pair. To determine the upfront premium for capping a ride's price at $K$ over a period $[0, T]$, we therefore need the joint dynamics of offers and intent-to-request conversion, which are not always available.

Calculating this premium requires computing (a) the ride initiation likelihood (conversion), (b) the probability that the uncapped price would have exceeded $K$, and (c) the present value of any potential price exceeding $K$. Available historical data on empirical ride initiation frequencies is statistically sufficient to compute (a). The sparsity of ride data necessitates a theoretical model of the price distribution to compute (b) adequately. Once (b) is determined, (c) can be computed using known financial tools. A generalized pricing distribution is, thus, at the core of achieving our objective of price capping.

## 1.3 Outline

Section 2 collects the necessary background: metric graphs (Sec. 2.1), cyclostationary processes over them (Sec. 2.2), and the resulting travel-time process (Sec. 2.3), which we cast as an arithmetic Brownian motion on the graph. Section 3 states the market assumptions under which we price. Section 4 develops the pricing equation: we first derive (Sec. 4.1) a

[2]Consider a business establishment, such as a gym or medical clinic, with repeat clients who visit on regular schedule. To establish a stronger business-client relation, the business is considering providing personalized transportation packages, where clients have the option to cap their trip cost to a predetermined amount, regardless of the actual cost.

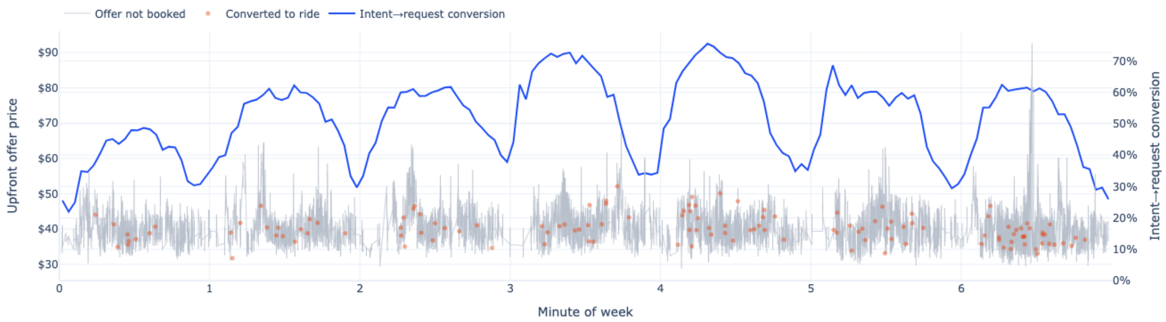


Figure 1: Offered prices (grey) and the sample of converted, paid prices (orange) over one week, together with the intent-to-request conversion rate. Location and time are anonymized for privacy.

closed-form premium for the on-demand price guarantee under both a population and a route-conditional travel-time law. Later (Sec. 4.2) we package this single premium into products of increasing informational difficulty, scheduled rides and memberships.

Because these premiums must be computed on sparse data, Section 5 introduces route and travel-time samplers, and Section 6 describes the 2014 Quebec City GPS data (QCD) on which we evaluate them. Section 7 is the empirical core: we verify that the samplers reproduce the out-of-sample trip-time distribution (Sec. 7.2), report the traffic-parameter estimates that feed the premium (Sec. 7.4), and study the guarantee's profit-and-loss behaviour as a function of the strike $K$ (Sec. 7.5), of a bundled discount (Sec. 7.6), and of deliberate membership abuse (Sections 7.7–7.8). Section 8 we discuss open ended problems and analysis.

### 1.4 Literature review

Previous research has shown that incorporating uncertainty into travel time estimates can lead to improved outcomes in ride-sharing systems, such as higher driver-rider match rates (up to 85%) and lower average prices (up to 12%) [Long et al., 2018]. These improvements can also positively impact other key system metrics, including reliability, service rate, utilization, and profitability [Li et al., 2021, 2022a,b]. Consequently, interest in understanding travel time uncertainty has been growing for practical reasons [Westgate et al., 2013, Wang et al., 2019a, Hunter et al., 2009, Westgate et al., 2016]. Researchers have explored various approaches, ranging from models that estimate uncertainty from individual edges [Zheng and J van Zuylen, 2013, Jenelius and Koutsopoulos, 2013] to those that estimate route and travel uncertainty simultaneously [Westgate et al., 2013, Wang et al., 2019a, Hunter et al., 2009, Westgate et al., 2016]. With the increasing volume of collected GPS data, research has primarily focused on understanding the distribution of travel time through parametric methods [Woodard et al., 2017, Guo et al., 2012, Ma et al., 2017, Zhang et al., 2019] or by examining the limiting properties of stochastic and dynamical processes resembling real-world travel behaviour Elmasri et al. [2023], Bolin et al. [2022].

Recent progress has shown that deviations in travel time behave similarly to deviations from a normal distribution Elmasri et al. [2023], Woodard et al. [2017]. Therefore, it is not surprising that regression-based models have shown promising results in travel time modelling [Westgate et al., 2016, Woodard et al., 2017, Budge et al., 2010, Westgate et al., 2013]. In particular, Elmasri et al. [2023] have suggested a Gaussian form as a population-style distribution for travel time, where both the mean and variance scale with distance. For example, a $N(n\mu, n\sigma^2)$, where $n$ is the number of road segments and $(\mu, \sigma^2)$ are map- (possibly traffic time bin-) specific mean and variance constants. A similar distribution has been proposed for route-conditional travel time.

For a survey of dynamical processes on networks (including transportation networks), see Barrat et al. [2008, ch. 11], for resent statistical work on the topic see [Kolaczyk and Csárdi, 2014, Snijders et al., 2017, Burk et al., 2007, Britton and O'Neill, 2002, Golightly and Wilkinson, 2005, Bolin et al., 2022].

Dynamic pricing is central to ride-sharing platforms, enabling real-time price adjustments that balance supply and demand while maintaining service quality [Banerjee et al., 2015, Yan et al., 2020]. Such pricing has been shown to increase driver availability, reduce peak-hour demand, and limit inefficient pickups [Chen et al., 2015, Hall et al., 2015], though price volatility and fairness remain open challenges [Yan et al., 2020]. We deliberately abstract from this two-sided, equilibrium view: our object is not the market-clearing price but the *cost* of a guarantee written on top of it, so we take the underlying fare process as given and price contracts against it.

Pricing such a guarantee is a contingent-claim problem. The idea of valuing a service guarantee as a financial derivative, rather than as a price to be optimized, has precedent outside transportation, for instance [Rasmusson, 2001] prices access to a virtual path across a communication network, over a future time interval and at a strike $K$, as a Black–Scholes-type option whose underlying is a stochastic process on network resources. Our construction is structurally analogous but instantiated on a road network. One caveat inherited from this analogy is that road travel time is not a traded, hedgeable asset, so the market is incomplete and no replicating portfolio exists. We therefore price the guarantee at its discounted expected cost under the modelled travel-time law (Section 3), a cost-based premium rather than a no-arbitrage replication price.

Beyond single rides, there is growing interest in pricing more complex transportation products, such as commuter memberships; most work here focuses on optimization methods for commuter ride-sharing [Wang et al., 2019b, Ma and Zhang, 2017]. A membership priced at a flat cap, however, raises a selection problem familiar from flat-rate and subscription pricing: a price paid regardless of usage attracts precisely the users for whom it is least profitable and weakens their incentive to economize [Della Vigna and Malmendier, 2006, Lambrecht and Skiera, 2006]. The distinction between this selection channel and the usage (moral-hazard) channel is well studied in insurance [Powell and Goldman, 2021], and it is the lens we apply to membership abuse in Sections 7.7–8.

# 2 Background

## 2.1 Metric graphs

A finite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a set of a finite set of vertices $\mathcal{V} = \{v_i\}$ and a set $\mathcal{E} = \{e_j\}$ of edges connecting the vertices. We write $u \sim v$ or $(u, v) \in E$ if $u$ and $v$ are connected with an with edge. We let $\mathcal{G}$ be a direct graph, that is, $(u, v)$ represents and edge from $u$ to $v$. We call a graph $\mathcal{G}$ a metric graph, if every edge $e$ is assigned a length $0 < l_e < \infty$, and over $e$ it possible to assign a coordinate $c \in [0, l_e]$ which increases in a specified (but otherwise arbitrary) direction along the edge. Since the length $l_e$ are assumed to be finite, this implies that the graph is compact. A point $s \in \mathcal{G}$ is a position on an edge, i.e. $s = (e, x)$, where $x \in [0, l_e]$. A path $\rho = \langle s_0, s_1 \rangle \in \mathcal{G}$ is a sequence of edges $(e_1, \ldots, e_n)$ where the path starts at point $s_0 \in e_1$ and ends at $s_1 \in e_n$. A natural choice of metric for the graph is the shortest path distance, which for any two points in $\mathcal{G}$, it is defined as the length of the shortest path in $\mathcal{G}$ connecting the two. We denote this metric by $\delta(\cdot, \cdot)$ from now on. We provide a path conditional version of this distance metric as $\delta_\rho(s) = \delta_\rho(s, s_0)$, for $s \in \rho$, as the distance between points $s$ and the start of the path $s_0$, over the the path. We further let $\delta_\rho$ defines the whole distance of $\rho$. Let $n_\rho = |\{e : e \in \rho\}|$ be the number of edges of $\rho$. We assume that the graph is connected, so that there is a path between all vertices. We define $\Pi(e_1, e_2)$ to be the set of all paths between edge $e_1$ and $e_2$, and similarly $\Pi(s_0, s_1)$ be the set of all paths between points $s_0, s_1 \in \mathcal{G}$. There is no unique way of defining process over graphs. This is a result of various way of defining the boundary condition (vertex condition), such as the Neumann-Kirchhoff condition, see Berkolaiko and Kuchment [2013, Eq (1.4.4) and Them 1.4.4]. In this work, we treat a path $\rho = \langle s_0, e_1, e_n, s_1 \rangle$, as an extended sequence of an uninterpreted edge, having the geometry

$$\left[0,\ \delta(s_0, e_1) + \sum_{e \in \rho} l_e + \delta(e_n, s_1)\right],$$

without introducing intermediate vertices at $l_e$, $e \in \rho$.

For every edge $e \in \mathcal{E}$, a function $f_e \in L_2(e)$ if $f_e$ is square-intergrable over the edge $e$. We further define a function $f$ over $\mathcal{G}$ as a collection of functions $f = \{f_e\}_{e \in cE} \in L_2(\mathcal{G})$ if $f_e \in L_2(e)$ for each $e \in \mathcal{E}$. Tihs results in $L_2(\mathcal{G})$ being the direct sum over $L_2(e)$ as

$$L_2(\mathcal{G}) = \bigoplus_{e \in \mathcal{E}} L_2(e).$$

We finally let $C(\mathcal{G}) = \{f \in L_2(\mathcal{G}) : f \text{ is continuous}\}$ be the space of continuous functions on $\mathcal{G}$. We define $\mathbb{R}_+$ to be the positive part of the real line.

## 2.2 Cyclostationary processes over metric graphs

We let $X = X(t, s)$, for every fixed time index $t \in \mathbb{R}_+$ and point $s \in \mathcal{G}$, be a 2nd order cyclostationary process in the wide sense. Here $X(t, s)$ is spatio-temporal process defined as

$$X(t, s) = m(t, s) + \epsilon(t, s), \qquad t \in \mathbb{R}_+, s \in \mathcal{G}, \tag{2.1}$$

where, in the temporal domain (holding the spatial fixed), for every $e \in \mathcal{E}$ and $t, u \in \mathbb{R}_+$, we have

$$m(t, s) = \mathbb{E}[X(t, s)] = m(t + T_e, s), \tag{2.2a}$$
$$\sigma^2(t, s) = \mathbb{E}[\epsilon^2(t, s)] = \sigma^2(t + T_e, s), \tag{2.2b}$$
$$\varsigma(t, u, s) = \mathbb{E}[\epsilon(t, s)\epsilon(u, s)] = \varsigma(t + T_e, u + T_e, s), \tag{2.2c}$$

where $T_e \in \mathbb{R}_+$ is an absolute positive constant defining the temporal cycle of edge $e$. Here, $m(t, s), \sigma^2(t, s), \varsigma(t, u, s) \in C(\mathcal{G})$ (measurable functions over $\mathcal{G}$), and $\mathbb{E}[\epsilon(t, s)] = 0$ for all pairs $(t, s) \in \mathbb{R}_+ \times \mathcal{G}$. We further let $\max_{e \in \mathcal{E}} T_e \le T_{\mathcal{G}} < \infty$ be the global temporal cycle over $\mathcal{G}$. That is, all equalities in (2.2) apply when replacing $T_e$ by $T_{\mathcal{G}}$. By definition of cyclostationarity, for come absolute constant $U_1 > 0$, we have

$$|m(t, s)| + \left|\sigma^2(t, s)\right| \le \max_{u \in [0, T_{\mathcal{G}}]} \left\{m(u, s), \sigma^2(u, s)\right\} \le U_1,$$

for all $t \in \mathbb{R}_+$, which is a stronger condition than the classical growth condition imposed on stochastic random variables, see Oksendal [2013, Thm. 5.2.1]. We further assume Lipschitz continuity in space and time as

$$|m(t, x) - m(u, y)| + \left|\sigma^2(t, x) - \sigma^2(u, y)\right| \le C_2 \left\{|t - u| + |x - y|\right\}, \tag{2.3}$$

for $t, u \in [0, T_{\mathcal{G}}]$, $x, y \in \mathcal{G}$, and $C_2 > 0$ is an absolute constant. (2.3) results in spatial Lipschitz continuity when $t = u$, and temporal when $x = y$.

We assume a conditional independence condition between space and time, that is is

$$X(t,x) \perp X(t,y) \mid t; \qquad x,y \in \mathcal{G}, x \neq y.$$

We define a ride on $\mathcal{G}$ to be a the pair $(\rho, t)$, representing the path $\rho$ of the ride and the start time $t_2 > 0$.

## 2.3 Travel time process

We first define a transportation network $\mathcal{N} = (\mathcal{G}, X)$ to be a metric graph $\mathcal{G}$ equipped with wide-sense cyclostationary process $X(t,s)$, $t \in \mathbb{R}_+$ and $s \in \mathcal{G}$. Travel time over a path $\rho = \langle s_0, s_1 \rangle \in \mathcal{G}$ can be directly computed as the stochastic integral

$$\mathcal{T}_\rho = \int_\rho X(\mathcal{T}_s, s)ds = \int_\rho m(\mathcal{T}_s, s)ds + \int_\rho \epsilon(\mathcal{T}_s, s)ds, \quad \mathcal{T}_{s_0} = 0 \tag{2.4}$$

where $\int_\rho$ represent integration over the predetermined path $\rho$, with $s \in \rho$ being a point over the path. Here, $\mathcal{T}_s, s \in \rho$ defines the travel time up to location $s$ on $\rho$, such that the travel time at the start point $s_0$, $\mathcal{T}_{s_0} = 0$. Elmasri et al. [2023] approximated the integral (2.4) by a discrete version $\mathcal{T}_\rho^d$, as

$$\mathcal{T}_\rho^d = \sum_{e \in \rho} m_e(\tau_e) + \epsilon_e(\tau_e),$$

where

$$m_e(t) = \int_e m(t,s)ds, \quad \sigma_e^2(t) = \int_e \sigma^2(t,s)ds, \quad \epsilon_e(t) = \int_e \epsilon(t,s)ds, \tag{2.5}$$

for $t \in \mathbb{R}_+$. (2.5) are the spatial marginals over $X(t,s)$ defined in (2.1). We have implicitly assumed that all edges are traversed in $\mathcal{T}_\rho^d$, and that $\mathcal{T}_\rho^d$ is defined on a transportation network $\mathcal{N}^d$ defined by the marginals of $X$, such that $\mathcal{N}^d = (\mathcal{G}, X^d)$, and $X^d = \{X_e(t)\}_{e \in \mathcal{E}}$, where $X_e(t) = \int_e X(t,s)ds$. The following lemma is an adaptation of Elmasri et al. [2023, Lem 6, Them. 6] that characterize the asymptotic distribution of $\mathcal{T}_\rho^d$.

**Lemma 2.1** (Population distribution Elmasri et al. [2023])**.** Consider $\rho$ as a random walk over the $\mathcal{N}^d$ and let $\tau = \{\tau_e\}_{e \in \rho}$ be the random arrival times over edges of the walk $\rho$. Let $\tau$ be sufficiently mixing[3]. Then as the number of edges in $\rho$ grow to infinity (i.e. $n_\rho \to \infty$), we have

$$n_\rho^{-1} \mathcal{T}_\rho^d \xrightarrow{\text{a.s.}} \mu \tag{2.6a}$$

$$n_\rho^{-1/2}(\mathcal{T}_\rho^d - n_\rho \mu) \xrightarrow{d} N(0, \sigma^2), \tag{2.6b}$$

where $\mu$ and $\sigma^2$ are absolute positive constants that are independent of initial conditions and of $\rho$.

The result in (2.6) suggest that deviations of $\mathcal{T}_\rho$ around the average travel time per edge $\mu$, normalized by $\sqrt{n_\rho}$, resemble independent samples from a normal distribution, with variance that is independent of the route. However, (2.6) is an asymptotic distribution over a population of rides in a system. That is for an arbitrary ride from a population of rides, the probability it arrives before $n_\rho \mu$ is 1/2. For a specific ride, say $(\rho_i, t_2)$, this probability can be significantly different if $\mathcal{T}_{\rho_i}$ has not reached the asymptotic distribution. The following lemma is an adaptation of Elmasri et al. [2023, Thm. 7] that characterized the distribution of travel

[3] at least $\rho$-mixing, see Bradley [2005] and Elmasri et al. [2023, Lem. 5,6]

time for a specific ride. First, we define a deterministic mean and covariance sequences for a ride over $\rho = \langle e_1, e_{n_\rho} \rangle$, starting at $t_1 \in \mathbb{R}_+$, as

$$t_{i+1} = t_i + m_{e_i}(t_i), \quad i \in \{2, \dots, n_\rho\}, \tag{2.7a}$$

$$\mu_\rho = \sum_{i=1}^{n_\rho} m_{e_i}(t_i), \tag{2.7b}$$

$$\sigma_\rho^2 = \sum_{i=1}^{n_\rho} \sigma_{e_i}^2(t_i) + 2\xi \sum_{i=2}^{n_\rho} \sigma_{e_i}(t_i)\sigma_{e_{i-1}}(t_{i-1}), \tag{2.7c}$$

for some absolute constant $\xi$. Here, $m_e$ and $\sigma_e^2$ are the spatial marginals defined in (2.5).

**Lemma 2.2** (Route-predictive distribution, Thm. 6 in Elmasri et al. [2023])**.** Under the conditions of Lemma 2.1, for an arbitrary start time $t_0 \in \mathbb{R}_+$, given a path $\rho = \langle e_1, e_{n_\rho} \rangle \in \mathcal{G}$, we have

$$\sigma_\rho^{-1} \left( \mathcal{T}_\rho^d - \mu_\rho \right) \xrightarrow{d} N(0, \eta), \quad \text{as } n_\rho \to \infty, \tag{2.8}$$

where $\eta$ is a strictly positive constant that is independent from initial conditions, and $\sigma_\rho^2$, $\mu_\rho$ are defined in (2.7).

The result of Lemma 2.2, although still suggests approximate normality of travel time distribution, it integrates route-specific information in the mean and covariance functions (2.7) to better approximate this distribution. As shown in Elmasri et al. [2023, Fig. 3], the approximation in (2.8) can achieve significant accuracy when compared to (2.6b).

Working with $\mathcal{T}_\rho$ in (2.4) directly is challenging, since the distribution of $\epsilon$ is not well understood. Every edge $e \in \rho$ can influence $\mathcal{T}_\rho$ differently, depending on the cyclostationarity pattern of the variance $\sigma(t, s)$ and the actual error distribution.

To overcome such challenges, we rely on the results in Lemma 2.1 and 2.2. The mean and covariance functionals $\mu(t, s)$, $\sigma(t, s)$ in (2.2) define manifolds over the graph $\mathcal{G}$. Using the asymptotic normality of travel and those functionals, we redefine travel time as the stochastic process, initiated at $s_0 = 0$, as

$$d\mathcal{T}_s = \alpha(\mathcal{T}_s, s)ds + \beta(\mathcal{T}_s, s)dB_s, \qquad \mathcal{T}_{s_0} = 0, \quad s \in \rho, \tag{2.9}$$

where $B_s$ is Brownian process. (2.9) suggests that travel time over a route $\rho$ behaves like an arithmetic Brownian motion on the spatial domain, having the expectation and variance as

$$\mathbb{E}[\mathcal{T}_s] = \int_\rho \alpha(\mathcal{T}_s, s)ds \quad \mathrm{Var}[\mathcal{T}_s] = \int_\rho \beta^2(\mathcal{T}_s, s)ds, \quad s \in \rho,$$

where the variance follows by the assumption that $B^2(t, s)$ is bounded, and thus by the Itô isometry property Oksendal [2013, Lem. 3.1.5], we have $\mathbb{E}\left(\int_\rho \beta(\mathcal{T}_s, s)dB_s\right)^2 = \int_\rho \beta^2(\mathcal{T}_s, s)ds$.

Finally, to adapt $\mathcal{T}_\rho$ to the population version in Lemma 2.1 and the route-specific version in Lemma 2.2, we let $\alpha$ and $\beta$ be

$$\alpha(t, s) = \begin{cases} \mu & \text{population} \\ m(t, s) & \text{route-specific} \end{cases} \qquad \beta(t, s) = \begin{cases} \sigma & \text{population} \\ \sqrt{\eta}\sigma(t, s) & \text{route-specific,} \end{cases} \tag{2.10}$$

where $m(t, s)$ and $\sigma(t, s)$ are as defined in (2.2) and $\eta$ is a universal parameter that is estimated from data and accounts for residual error.

# 3 Assumptions in ride-share markets

In traditional pricing theory, three primary approaches dominate: cost-based, market-based (competition-driven), and value-based pricing Phillips [2021]. This study proposes a pricing policy for a ride-sharing platform that is entirely cost-based, independent of real-time supply-demand dynamics. Unlike prior research focusing on matching algorithms or market-based pricing (see 1.4), we assume prices are determined by equation (4.1). The goal is to recover operational costs based on traffic and route assumptions, rather than to maximize profit. We operate in a monopoly market, assuming all ride requests are satisfied immediately (no unmatched requests).

Our analysis focuses on the pricing mechanism from the moment a rider submits a request until the driver completes the trip. To isolate the pricing dynamics, we make the following assumptions to focus on the pricing mechanism:

($A_1$) Once a rider submits a request, they do not cancel, and a driver accepts and completes the trip immediately, with no rematching or wait time.

($A_2$) The platform is a monopolist with full pricing power, facing no competition or alternative transportation options for riders.

($A_3$) For every ride request, an available driver exists to fulfill it within a finite time.

($A_4$) Both riders and drivers are non-strategic and price-insensitive beyond the modelled price; they do not respond to changes in market conditions or alternative routing options.

($A_5$) No ride-specific information (e.g., route, traffic, or weather) is known at the time of membership subscription. (Once they hold the membership, they may abuse it.)

Moreover, we assume a frictionless environment:

($B_1$) The risk-free interest rate $r$ is constant over time and known.

($B_2$) There are no transaction costs, taxes, or additional platform fees beyond the modeled price.

Under our assumptions, riders are price-insensitive but may increase usage in response to external conditions (i.e, abuse as mentioned in section 7.7). This setup reflects a monopoly market with non-strategic supply, where the platform acts as the sole decision-maker, and the customer accepts the offered price without negotiation or cancellation. The pricing method is not explicitly adjusted based on real-time fluctuations in driver supply or rider demand.

Our objective is to propose a cost-based pricing method and evaluate its stability and volatility under dynamic conditions, such as varying traffic or membership abuse, without modeling equilibrium outcomes in a two-sided market. We explicitly exclude the matching process, strategic driver behavior, and real-time supply-demand equilibrium mechanisms from our analysis.

While dynamic pricing based on real-time demand and supply can, in theory, maximize revenue, it is often operationally costly and computationally intensive. In practice, rider behavior is highly unpredictable and sensitive to wait time and price fluctuations, leading to instability. The tradeoff in our method is a potential loss in revenue in exchange for greater pricing stability and predictability.

# 4 Pricing equation in ride-share markets

Ride pricing is determined by three constant factors predefined by the service provider. A fixed price $C_0$ (starting price), a cost for a unit of time $C_1$, and a cost for a unit of distance $C_2$. Let $P_\rho(s,t)$ be the price of a ride that starts at time $t > 0$, over the route $\rho$, and at location $s \in \rho$, that is

$$P_\rho(s,t) = C_0 + C_1 \mathcal{T}_s(t) + C_2 \delta_\rho(s), \qquad s \in \rho, \tag{4.1}$$

where $\mathcal{T}_s(t) = \mathcal{T}_{\rho(s)}(t)$ is the travel time up to point $s \in \rho$ and starting at time $t > 0$ and $\delta_\rho(s) = \delta(s, s_0)$ is the length of the path $\rho$ up to point $s$. To simplify notations, we will drop the $(t)$ from $\mathcal{T}_s(t)$, since the start time defined in the pricing function $P_\rho(s,t)$. The base price is determined at the start point $s_0 \in \rho$, that is

$$P_{s_0}(s_0, t) = C_0, \quad \text{for all } t > 0,$$

and the final price is determined at the destination point $s_1$, and arrival time $t_2$, as

$$P_\rho(s_1, t) = C_0 + C_1 (t_2 - t) + C_2 \delta_\rho.$$

Here, $\mathcal{T}_{\rho(s_1)} = t_2 - t$, an actual realization. Otherwise, for any point $s \in \rho \setminus \{s_0, s_1\}$, the price equation (4.1) is a random variable determined by $\mathcal{T}_s(t)$. Moreover, for any two points $s_1, s_2 \in \rho$, if $s_2$ is further along $\rho$ than $s_2$ (i.e. $\delta_\rho(s_1) \leq \delta_\rho(s_2)$), then

$$P_\rho(s_2, t) \geq P_\rho(s_1, t).$$

Such inequality in the time domain is not true, that is $P_\rho(s, t_1)$ is not necessary less than $P_\rho(s, t_0)$ for $t_1 < t_0$, and $s \in \rho$. The price equation (4.1) is, thus, composed of three components: (i) the fixed price $P_\rho(0,0) = C_0$, (ii) the incremental price $C_2\delta_\rho(s)$, and (iii) the variable price $C_1\mathcal{T}_s$, $s \in \rho$. In this sense, the true price of a ride over $(t, \rho)$ that arrived at $\mathcal{T}_\rho = t_2 - t$, is

$$P_\rho(s_1, t) = C_0 + C_1(t_0 - t) + C_2\delta_\rho, \tag{4.2}$$

If the realization is not yet observed, then the present value of the ride price at start location $s_0$ is a random variable having the expectation

$$P_\rho(s_0, t) = \mathbb{E}\left[P_\rho(s_1, t)\right] = C_0 + C_1 \mathbb{E}\left[\mathcal{T}_\rho\right] + C_2\delta_\rho, \tag{4.3}$$

In (4.3), the expectation is taken over all possible realization of $\mathcal{T}_\rho$, without accounting for the value of money. For example, having the option to allocate the cost of the ride, say $K$, to an investment vehicle with a risk-free annualized interest rate $i$, would earn $Ke^{i\mathcal{T}_\rho}$ by the time the ride arrives. In this sense, the price of a ride $(t, \rho)$ at the start location $s_0$ is

$$P_\rho(s_0, t) = \mathbb{E}\left[e^{-i\mathcal{T}_\rho} P_\rho(s_1, t)\right].$$

Adjusting for the value of money is mostly mathematical exercise, since in most real-world applications, arrival times are within minutes. Exceptions exists, for example cargo and fright, which can take last weeks. Such adjustment can be useful for schedules of rides or membership pricing as shown in later sections.

Finally, the present value of the pricing formula can be extended to route unconditional pricing through expectations. Given the set of all path $\Pi(s_0, s_1)$ between origin-destination pair $(s_0, s_1)$. The price of the ride $(t, s_0, s_1)$ is

$$P_{(s_0,s_1)}(s_0, t) = \mathbb{E}\left[e^{-i\mathcal{T}_{\rho_k}} P_{\rho_k}(s_1, t)\right], \qquad \rho_k \sim \Pi(s_0, s_1),$$

where the expectation is taken over the uniform distribution of routes, and travel time. A discrete probability measure can be applied on the set $\Pi$, for a non-uniform distribution.

## 4.1 Up-front pricing

Up-front pricing, a popular initiative by ride-share providers, promises a price $K$ at the request time $t$ for a ride over the path $\rho$ starting at location and time $(s_0, t_0)$, $t_0 \geq t$, and potentially ending a location and time $(s_1, t_1)$. When $t_0 = t$, we call this an on-demand ride, otherwise this is a scheduled ride. The progression if a ride request is shown in Figure 2.

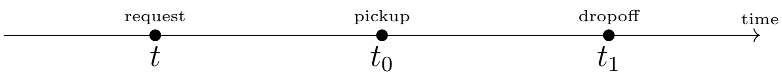


Figure 2: Time progression of a rides request, pickup and dropoff.

The upfront price $K$ is guaranteed unless the realized price $P_\rho(s_1, t_1)$ does not exceed a cutoff threshold, which is often defined as a percentage of the upfront price, say $Ke^\zeta$, for a predefined constant $\zeta \in \mathbb{R}_+$ predefined[4]. Under Assumption (B$_1$), a request if accepted is always fulfilled. let $R(t_1, t_0, \rho, K)$ be the cost of this liability at the time of execution (i.e. the arrival time $t_1$), defined as

$$R(t_1, t_0, \rho, K) = (P_\rho(s_1, t_1) - K)^+ \mathbf{1}_{\left\{P_\rho(s_1,t_1) \geq Ke^\zeta\right\}}. \tag{4.4}$$

In this sense, the final price charged to the customer at time $t_1$ is

$$K + R(t_1, t_0, \rho, K) = \begin{cases} P_\rho(s_1, t_1) & \text{if } Ke^\zeta \leq P_\rho(s_1, t_1) \\ K & \text{otherwise.} \end{cases} \tag{4.5}$$

It is clear from (4.4) that for promised ride price $K_1 \geq K_2$, we have

$$R(t_1, t_0, \rho, K_1) \leq R(t_1, t_0, , \rho, K_2).$$

Oftentimes rideshare providers do not explicitly estimate the cost of such a guarantee at time the request time $t$, that is $R(t, t_0, \rho, K)$. Instead, consumers are only charged $K\$$ at time $t_0$. At arrival time $t_1$, the price is adjusted according to (4.5).

Under Assumptions 3, we model the travel time along the path $\rho = \langle s_0, s_1 \rangle$ as an asymptotic stochastic dynamic governed by the partial differential equation (2.9). The price of the guarantee (4.4) at request time $t$ is then given by:

$$\begin{aligned} R(t, t_0, \rho, K) &= \mathbb{E}\left[e^{-r(t_0-t)} R(t_1, t_0, \rho, K)\right] \\ &= \mathbb{E}\left[e^{-r(t_0-t)-r\mathcal{T}_\rho} (P_\rho(s_1, t_1) - K)^+ \mathbf{1}_{\left\{P_\rho(s_1,t_1) \geq Ke^\zeta\right\}}\right] \\ &= e^{-r(t_0-t)} e^{-rd_7} \left[\left\{d_0 - rC_1 d_1^2 - K\right\} \{1 - \Phi(d_5)\} + C_1 d_1 \phi(d_5)\right] \end{aligned} \tag{4.6}$$

where

$$\begin{aligned} d_0 &= C_0 + C_1 \mathbb{E}[\mathcal{T}_\rho] + C_2 \delta_\rho \\ d_1 &= \{\mathrm{Var}[\mathcal{T}_\rho]\}^{1/2} \\ d_4 &= \frac{Ke^\zeta - d_0}{C_1 d_1} \\ d_5 &= d_4 + r d_1 \\ d_7 &= \mathbb{E}[\mathcal{T}_\rho] - r d_1^2/2 \end{aligned}$$

[4] a link to pricing policy to be added

When $r = 0$, the solution to (4.6) simplifies to

$$R(t, t_0, \rho, K) = (d_0 - K)\{1 - \Phi(d_5)\} + C_1 d_1 \phi(d_5).$$

Further if $\zeta = 0$, as there is no minimum threshold for the guarantee, we have

$$R(t, t_0, \rho, K) = (d_0 - K)\{1 - \Phi(d_8)\} + C_1 d_1 \phi(d_8),$$

where

$$d_8 = \frac{K - d_0}{C_1 d_1}.$$

The next section introduces more complex pricing structures with unknown routes and start times, and packages them into membership offering.

## 4.2 Pricing complex products

Pricing of on-demand single rides in Section 4 is done under the simplest case of known (route, start time)= $(\rho, t_0)$. We vary these two quantities by generalizing from a known fixed route to only the (start, end) = $(s_0, s_1)$ locations are known, and then an unknown start time. Summarized in three pricing products

($P_1$) On-demand. An instantaneous ride request to go from $s_0$ to $s_1$ in $\mathcal{G}$ over the route $\rho$.

($P_2$) Scheduled. A ride request to go from $s_0$ to $s_1$ in $\mathcal{G}$ at a future time $t$.

($P_3$) Membership. An option to request up to $M$ rides within a pacified time interval, $T_M \in \mathbb{R}_+$, at a predetermined price $K$. Such rides can be (but not necessarily) conditioned over fixed hours and locations.

Computing an accurate upfront price–paid at the time of request or subscription–is the central challenge in three scenarios of increasing complexity: ($P_1$), ($P_2$), and ($P_3$). The information available at the pricing time diminishes with each case. In the on-demand scenario ($P_1$), both the route's start/end points and the real-time network dynamics are known. For a scheduled ride ($P_2$), only the start/end locations are known, requiring a forecast of network dynamics. Finally, under a membership plan ($P_3$), no details are known at the pricing stage, and the ride itself is elective.

The premium price of the on-demand ($P_1$) scenario is defined in (4.6). Recall that $\Pi(s_0, s_1)$ is the set of routes between any two points, $s_0$ and $s_1$. Let $\pi(\rho)$ be the likelihood of route $\rho$ in $\Pi(s_0, s_1)$. If a uniform distribution is desired, then $\pi(\rho) \propto 1/|\Pi(s_0, s_1)|$. $\pi(\rho)$ can (not necessarily) time dependent. The premium price of the scheduled ride ($P_2$) requested time $t$ for (start time, start location, end location) = $(t_0, s_0, s_1)$ is

$$\begin{aligned}
\mathsf{R}_{\mathsf{sch}}(t, t_0, K) &= \sum_{\rho_i \in \Pi(s_0, s_1)} R(t, t_0, \rho_i, K)\pi(\rho_i) \\
&= \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E}\left[e^{-r(t_0 - t + \mathcal{T}_{\rho_i})} R(t_1, t_0, \rho, K)\right]\pi(\rho_i) \\
&= \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E}\left[\mathbb{E}\left[e^{-r(t_0 - t) - r\mathcal{T}_{\rho_i}} R(t_1, t_0, \rho, K) \,|\, t_0\right]\right]\pi(\rho_i) \\
&= e^{-r(t_0 - t)} \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E}\left[e^{-r\mathcal{T}_{\rho_i}} R(t_1, t_0, \rho, K)\right]\pi(\rho_i).
\end{aligned} \tag{4.7}$$

The pricing in (4.7) occurs by averaging over all routes between $s_0$ and $s_1$, hence the notation $\bar{R}$. In the same spirit, let $\pi(t)$ be the distribution of pickup times on the interval

$T_R$. The price guarantee of a membership (P$_3$) purchased at time $t$ of $M$ rides on the interval ride (P$_2$), for fixed price $K$, is

$$
\begin{aligned}
\mathsf{R}_{\mathsf{mem}}(t, M) &= M \sum_{t_0 \in [0, T_R]} \sum_{\rho_i \in \Pi(s_0, s_1)} R\left(t_0, t_0, \rho_i, K_{\rho_i}\right) \pi(t_0) \pi(\rho_i), \\
&= M \sum_{t_0 \in T_R} e^{-r(t_0 - t)} \sum_{\rho_i \in \Pi(s_0, s_1)} R\left(t_0, t_0, \rho_i, K\right) \pi(t_0) \pi(\rho_i), \\
&= M \sum_{t_0 \in T_R} e^{-r(t_0 - t)} \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E}\left[e^{-r\mathcal{T}_{\rho_i}} R\left(t_1, t_0, \rho_i, K\right)\right] \pi(t_0) \pi(\rho_i), \\
&= M \sum_{t_0 \in T_R} e^{-r(t_0 - t)} \mathsf{R}_{\mathsf{sch}}\left(t_0, t_0, K\right) \pi(t_0),
\end{aligned}
\tag{4.8}
$$

where $\mathsf{R}_{\mathsf{sch}}$ follows from (4.7). To validate our method, the next section develops a statistical sampling methods for routes and travel time.

# 5 Statistical sampling of travel times and routes

To simulate alternative routes to a realized trip, we develop a statistical approach for trip sampling, and demonstrate its statistical fidelity. Given a trip (route, start time)$= (\rho, t_2)$, we alternative routes, then we sample travel times.

## 5.1 Route sampling

When all traffic states of the graph is observed, routing-finding algorithm, such as the one proposed by Yen [1970], can be used to list all routes from two points $(e_1, e_n)$ on the $\mathcal{G}$ at the desired start time $t_2$. Marrying this route sampling approach to reality is intuitive, but not always possible. Existing traffic data do not cover all states of the graph. Our objective is propose a sample method that provide statistically equivalent routes, that are graph-invariant for all triplet (start location, end location, start time) $= (e_1, e_n, t_2)$.

Given a route $\langle e_1, e_2 \ldots, e_n \rangle$, first, we match edges $\times$ time bins using a mean matching method. or each edge $e \in E$ and time bin $b \in B$, let $n_{e,b}$ be the sample size used to calculate the sample mean $\widehat{m}_e(t_b)$ and variance $\widehat{\sigma}^2_e(t_b)$. We define statistically equivalent edges by the difference between their means, as

$$
E_{e,b}(\zeta) = \left\{ (e', b') \in E \times B : \frac{\left|\widehat{m}_e\left(b\right) - \widehat{m}_{e'}\left(b'\right)\right|}{\sqrt{\frac{\widehat{\sigma}^2_e(b)}{n_{e,b}} + \frac{\widehat{\sigma}^2_{e'}(b')}{n_{e',b'}}}} \leq \zeta \right\}
\tag{5.1}
$$

The set in (5.1) is constructed using simultaneous t-tests, with unequal variance and sample size, on the hypothesis that the two edges $e, e'$ have equivalent means under the time bins $b, b'$. The critical value $\zeta$ can be chosen to achieve a $(1-\beta)100\%$, $\beta \in (0, 1)$ level of significance of the Student's t distribution with degrees of freedom following the Welch–Satterthwaite equation [Satterthwaite, 1946, Welch, 1947], as

$$
\text{degrees of freedom} = \frac{\left(\frac{\widehat{\sigma}^2_e(b)}{n_{e,b}} + \frac{\widehat{\sigma}^2_{e'}(b')}{n_{e',b'}}\right)^2}{\frac{\left(\widehat{\sigma}^2_e(b)/n_{e,b}\right)^2}{n_{e,b}-1} + \frac{\left(\widehat{\sigma}^2_{e'}(b')/n_{e',b'}\right)^2}{n_{e',b'}-1}}
$$

Second, we randomize the length of the sampled route using a normal distribution centered at the number of edges $n$ of a given realized route. This results in a route sampling process as

$$\begin{aligned} t_0 &\sim \textsf{given start time} \\ \rho = \langle e_1, e_2 \dots, e_n \rangle &\sim \textsf{given route } \rho \\ E_\rho &= \{E_{e,b} : (e,b) \in \rho\} \\ n' &\sim \lfloor \textsf{Normal}\,[n, \tau]\,\rfloor \\ \rho' = \langle e'_1, e_2 \dots, e'_{n'} \rangle &\sim \textsf{Uniform}\,[E_\rho]\,. \end{aligned} \tag{5.2}$$

Here $\lfloor Z \rfloor$ represent the integer floor of a positive variable $Z$.

## 5.2 Travel time sampling

We define $B$ to be the set of time bins of the data, for example, EveningRush, WeekDay, MorningRush, EveningNight, and WeekendDay, used in Elmasri et al. [2023]. For every edge $e \in \mathcal{G}$, let $\widehat{m}_e(t_b)$ and $\widehat{\sigma}_e(t_b)$ be the sample average speed and standard deviation of duration to travel $\mathcal{T}_e$, respectively. Here $t_b \in B$, meaning, i.e. a time bin. We reserve the notation $t_b$ to represent the time bin $t$ falls into. This results in $|E| \times |B|$ number of traffic states in $\mathcal{G}$.

Following the simple auto-regressive structure and the exact sampler proposed by Falk [1999], we sample route-specific travel time $\mathcal{T}_\rho$ as

$$\begin{aligned} t_0 &\sim \textsf{given start time} \\ \langle e_1, e_2 \dots, e_n \rangle &\sim \textsf{given route } \rho \\ \xi &\in (-1, 1) \;\; \textsf{a correlation coefficient} \\ (\Sigma)_{ij} &= \xi^{|i-j|}, 1 \le i, j \le n \\ (Z_1, \dots, Z_n) &\sim \textsf{multivariate-Normal}\,[0, 2\sin(\pi\Sigma/6)] \\ (U_1, U_1, \dots, U_n) &= (\Phi\{Z_1\}, \dots, \Phi\{Z_n\}) \\ \mathcal{T}_{e_i} &\sim \textsf{log-Normal}\,[\widehat{m}_e(t_{b_i}), \widehat{\sigma}_e(t_{b_i})] \\ t_{b_{i+1}} &= t_{b_{i-1}} + \mathcal{T}_{e_i} \\ \mathcal{T}_\rho &= \sum_{i=1}^{n_\rho} \mathcal{T}_{e_i}. \end{aligned} \tag{5.3}$$

The sinusoidal transformation of $\Sigma$ insures its positive definiteness, otherwise the nearest positive definite correlation matrix can used Higham [2002]. The sample mean and standard deviations are used, since the population version is rarely accessible in real-world applications.

The sampling method (5.3), is iterative. The travel times for the first edge at the start-time traffic-bin is sampled, and iteratively, the travel time of the second edge at the traffic bin of the start-time plus the travel time of the first edge is sampled, and so on until the last edge. This insures an $\alpha$-mixing fully-dependent. A consequence of the constructed dependence in the uniform series of random variables $(U_1, \dots, U_n)$. A non-iterative, route-invariant, population version of this sampling process is

$$\begin{aligned} t_0 &\sim \textsf{given start time} \\ \langle e_1, e_2 \dots, e_{n_\rho} \rangle &\sim \textsf{given route } \rho \\ \mathcal{T}_\rho &= n_\rho \widehat{\mu}. \end{aligned} \tag{5.4}$$

When the population mean is not accessible, we replace it by the sample mean $\widehat{\mu}$. For more details how to estimate $\widehat{\mu}$ refer to [Elmasri et al., 2023, Sec. 3.3.2].

# 6 Quebec City trip data

## 6.1 Data collection

Quebec City 2014 GPS data (QCD) is collected using the *Mon Trajet* smartphone application developed by Brisk Synergies Inc. This study uses a sample of open data,[5] which contained 21,872 individual trips. The sample contains no data that can be linked to individual drivers. While no data was collected during the winter months, the precise duration of the collection period is kept confidential. The application was installed voluntarily by over 4,000 drivers, who then anonymously logged information using a simple interface. The exact number of drivers is kept confidential.

## 6.2 Data cleaning

No measure was provided to ensure the validity of trips, i.e. if they were made solely by motor vehicles and not walkers or cyclists, and excluded non-traffic interruptions such as parking. Data is thus cleaned by breaking down long trips if there was more than 4 minutes of idle time or over 2 minutes between consecutive GPS observations. The start and end points of trips were trimmed. A trip officially began when the vehicle speed first exceeded 10 km/h and ended when it last dropped below 10 km/h. Trips were excluded if their median speed was less than 20 km/h, their maximum speed was less than 35 km/h, or their total driving distance (calculated from GPS points) was less than 1 km. After cleaning, the dataset contained 19,967 trips.

The median trip duration was 19 minutes (average 21 minutes), with the longest trip being 3 hours and 27 minutes. The median trip distance was 14.5 km (average 16.6 km), with a maximum of 170.4 km. The median time between consecutive GPS points was 4 seconds (average 9 seconds). Refer to [Elmasri et al., 2023, Sec.6.3] for more details on the cleaning process.

## 6.3 Map matching

A third-party service (TraxMatching[6]) was used to map trips' GPS observations to the Quebec City road network mapped by The OpenStreetMap Project[7] (OSM), a publicly accessible open-source project. This process is called map-matching, and numerous high-quality methods are available to do this [Newson and Krumm, 2009, Hunter et al., 2013]. For each trip, the third-party service returns a sequence of mapped GPS points with lengths equal to the original sequence. Each mapped GPS point is associated with a source "node id", "way id" and destination "node id" corresponding to a unique directional edge whose "way id" is between the source and destination nodes. The map-matching process resulted in 46,386 unique directional edges, which constitute the travelled portion of Quebec City, not its entirety. The average edge length is 170 meters, and the median is 81 m.

## 6.4 Traffic estimation

We estimate the total travel time per edge by calculating i) within-edge travel time, as the time spent within the edge, and ii) across-edge travel time, as the time spent between the two closest GPS observations, where one is in the edge and the other is in an adjacent edge. We calculate the across-edge distance in the same way. The total travel time per edge is

[5] A cleaned sample of the data is available at `https://github.com/melmasri/traveltimeCLT`.

[6] `https://www.traxmatching.io`

[7] `https://www.openstreetmap.org`

then 100% of within-edge plus across-edge travel time, weighted by half the proportion of across-edge distance to the total length of the edge. Total edge lengths are obtained from OSM. In rare circumstances, the map-matching service also returns intermediate edges that do not have initial GPS observations. This happens, for example, when a vehicle is moving fast or through a tunnel. We treat those intermediate edges, those without GPS observations, as a single edge and calculate the total travel time over it, and then assign edge travel time proportionally to the length of each intermediate edge. With these total travel time estimates, we calculate the average speed per edge by dividing the total travel time by the total length.

Figure 3 shows seasonal (weekly) traffic patterns per week hour, starting at the first hour of Sunday. The volume of traffic is reduced overnight on weekdays starting after 7 p.m. and on weekends. Daily traffic peaks are associated with a.m. and p.m. rush hours, with strong dips in between.

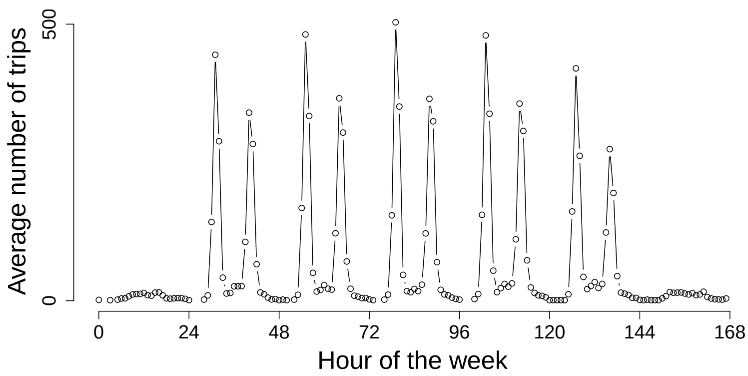


Figure 3: Average number of trips per hour in QCD, by hour of the week.

### 6.5 Data splitting

Because of the seasonal (weekly) traffic pattern illustrated in Figure 3 and the sparsity of GPS data for each edge, we classify our traffic data into three traffic time-bins (traffic-bins): i) an "AM-rush hour" bin for weekdays 6:30–8:30AM; ii) a "PM-rush hour" bin for weekdays 3:30–5:00 p.m., and iii) a "Non-rush hour" bin for all remaining periods. Alternative traffic-bins have been tested, but we found that aggregating data into these bins yielded the best results. In QCD, 37% of trips occurred in an AM-rush hour, with the same proportion for the PM-rush hour, and 26% in all other periods.

A test set of 2,000 trips is drawn uniformly at random without replacement, leaving the remaining 21,054 trips as training data. We classify a trip into a bin if all trip-edges are travelled within that bin; otherwise, the speed data for those trips are used for traffic estimation, but not trip analysis. Because the draw is unstratified, an edge×traffic-bin state observed only in a held-out trip has no training observation; such states fall back to the pooled "Global" bin at prediction time.

## 7 Data analysis

### 7.1 Outline

We evaluate the pricing framework of Sections 4–4.2 on the Quebec City data (QCD, Section 6) along the three products of increasing informational difficulty introduced in Section 4.2: on-demand rides ($P_1$) (route and start time known), scheduled rides ($P_2$) (only origin, destination, and start time known), and memberships ($P_3$) (no ride information at subscription).

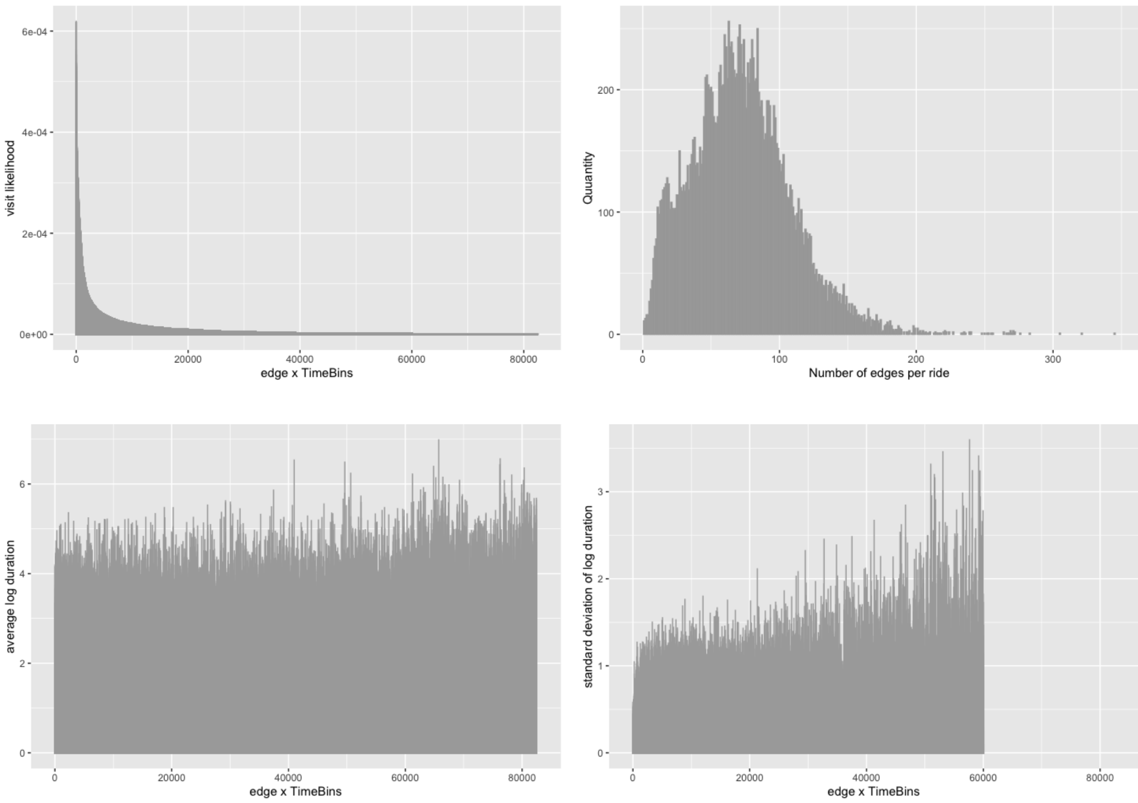


Figure 4: Descriptive statistics of the edge $\times$ time-bin representation in QCD. (Top left) Empirical visit likelihood of edge $\times$ time-bin states, sorted in decreasing order: a small fraction of states accounts for most observed traffic, motivating the frequency-weighted edge sampling in (5.2). (Top right) Distribution of the number of edges per trip, supporting the route-length randomization Normal$[n, \tau]$ in (5.2) and the large-$n_\rho$ regime of Lemmas 2.1–2.2. (Bottom left) Average log travel time per state, approximately flat across states. (Bottom right) Standard deviation of log travel time per state, which grows for rarer (higher-index) states with smaller sample sizes.

We first establish that the travel-time and route samplers of Section 5 reproduce the empirical trip distribution (Section 7.2), then report the traffic-parameter estimates that feed the pricing formula, and finally study the guarantee premium (4.6) and its profit-and-loss behaviour as a function of the strike price $K$, of a bundled discount, and of deliberate membership abuse.

Throughout, prices use the QCD tariff $C_0 = 3.17$ CAD, $C_1 = 0.31$ CAD/min and $C_2 = 0.9$ CAD/km, the risk-free rate is set to $r = 0$, and, unless stated otherwise, the strike is set to the expected price at the start of the ride, $K = \widehat{\mathbb{E}}\left[P_\rho(s_0, t_0)\right]$. The expectation $\mathbb{E}\left[\mathcal{T}_\rho\right]$ and variance $\mathrm{Var}\left[\mathcal{T}_\rho\right]$ entering (4.6) are estimated by the two models of Section 2.3: a *trip-specific* model that conditions on the route through the edge-level mean and variance functionals (Lemma 2.2) and a route-invariant *population* model (Lemma 2.1).

## 7.2 Fidelity of the sampling process

To justify pricing on simulated rather than observed rides, we first verify that the samplers of Section 5 reproduce the empirical trip-time distribution out of sample. We hold out 2,000 test trips, fit the edge $\times$ time-bin statistics on the remaining training trips, and, for each test trip, draw 1,000 travel times from each sampler and average them into a single trip estimate; the population sampler (5.4) requires no averaging and is drawn once per trip. We compare the full-order dependent sampler (5.3) against three reduced-dependence variants (independent, $\xi = 0$; first-order; second-order), the population sampler, and the two-state Hidden Markov model of [Woodard et al., 2017, Algo. 2]; the correlation parameter is fixed at $\xi = 0.31$ [Elmasri et al., 2023, Tab. 1].

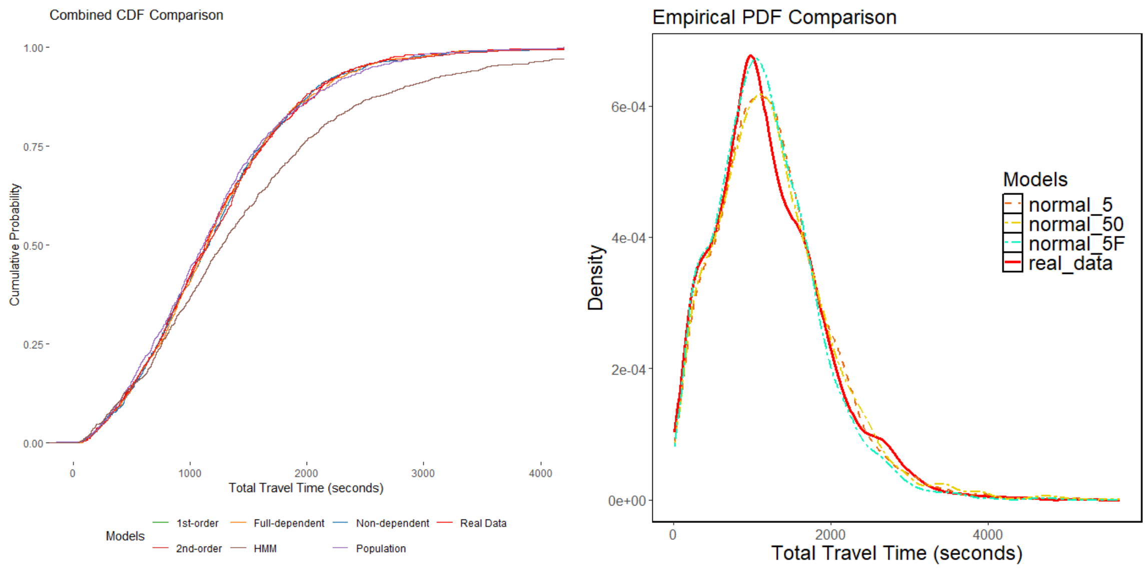


Figure 5: (Left) Comparing the empirical cumulative distribution of the 2000 test trips, our proposed travel time sampling methods, and the HMM model of [Woodard et al., 2017, Algo. 2]. (Right) Route-varied travel time simulation via (5.2) under normal and $t$-based edge matching at significance levels 0.05, 0.50, and 0.95.

Figure 5(left) and the full-order detail in Figure 6 show that all of our samplers track the empirical cumulative distribution closely, with fidelity increasing in the order of the retained dependence: the full-order sampler is best, followed by the second- and first-order variants. The population sampler, which discards serial dependence between consecutive edges, is visibly more dispersed but remains acceptable. Every one of our samplers improves substantially on the HMM alternative, which systematically overstates trip duration across the support. The marginal gain from adding higher-order dependence beyond the independent draw is small, indicating that most of the route-level uncertainty is already captured by the edge $\times$ time-bin marginals together with the log-normal edge model.

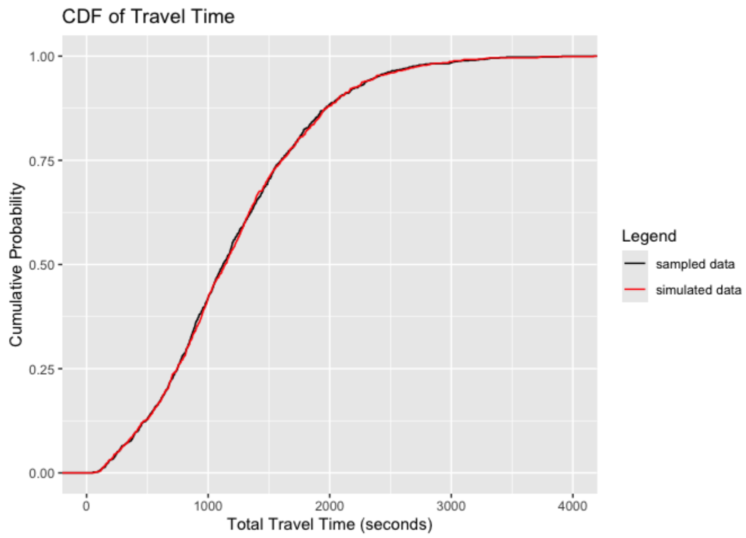


Figure 6: Empirical CDF of realized versus simulated trip travel time using the full-order dependent sampler (5.3) over 1000 QCD trips; the two distributions are nearly indistinguishable.

Route variation (5.2) preserves this fidelity: resampling each test trip into five statistically equivalent routes (matched by the simultaneous $t$-tests of (5.1)) and averaging their sampled travel times reproduces the empirical curve under normal edge matching at the 5% significance level (Figure 5, right); loosening the similarity threshold degrades the match, as expected.

Fidelity in travel time does not carry over to distance, and the gap matters for pricing. Edges are matched on the mean and variance of their log travel time, not on their length, and short edges are far more numerous than long ones at any given travel time, so the matching draws systematically shorter edges. On QCD the sampled routes carry the same number of edges as the trips they replace (a ratio of 0.99) but only 79% of their length on average (74% at the median). Since the fare (4.4) charges distance separately from time, a premium computed on observed routes and evaluated against sampled ones books that 21% as provider profit. The membership analyses below therefore price and realize in the same route space; a length-aware matching criterion in (5.1) would remove the discrepancy at the source and is left to future work. The descriptive statistics in Figure 4 explain why the procedure is well posed on QCD: a small fraction of edge $\times$ time-bin states accounts for most observed traffic (motivating the frequency-weighted edge sampling), the per-trip edge counts are large enough to support the asymptotics of Lemmas 2.1–2.2, and the per-state mean log travel time is roughly flat while its standard deviation grows only for the rarer, less-sampled states.

## 7.3 Profit-and-loss accounting

For a ride priced at strike $K$ with collected premium $R = R(t, t_0, \rho, K)$ from (4.6) and realized price $P = P_\rho(s_1, t_1)$, the provider's per-ride profit under cutoff $\zeta$ is

$$\mathsf{profit} \;=\; R \;-\; (P-K)^{+}\, \mathbf{1}_{\{P \geq K e^{\zeta}\}}, \tag{7.1}$$

i.e. the premium kept minus the payout triggered only when the realized price exceeds the guarantee threshold. We summarize a population of rides with five quantities: the share of profitable rides (*win rate*), the *profit factor* (total gains divided by the absolute total of losses, so values above one indicate net profitability), the *average profit* per ride, the *maximum single-ride loss*, and the average premium expressed as a percentage of the trip price, $R/P$. The premium ratio is the headline quantity for the rider, the profit factor and maximum loss for the provider.

## 7.4 Traffic-parameter estimation

The population model needs only the pair $\{\mu, \sigma\}$; we use the unbiased sample mean $\widehat{\mu}$ and the profile estimator $\widehat{\sigma}_{\text{prof}}$ of [Elmasri et al., 2023, Sec. 3.3.2], computed from 1,000 trips. Unlike $\widehat{\mu}$, which is stable across subsamples, $\widehat{\sigma}_{\text{prof}}$ is not: it varies by a factor of nearly eight across random draws of 500 trips from the same corpus, for the reason set out beneath Table 1. The trip-specific model additionally needs the edge-level pairs $\{\widehat{m}_e, \widehat{\sigma}_e^2\}$, the correlation $\xi$ and the residual variance $\nu$ of Lemma 2.2, estimated from 4,000 trips. Table 1 reports all estimates, both pooled ("at random") and stratified by traffic regime. Rush-hour travel is both slower and markedly less predictable than off-peak travel: the mean edge duration rises from 13.70 to 17.34 s and its standard deviation from 18.44 to 26.91 s, while the residual variance roughly doubles (1.15 vs. 2.04). The edge-to-edge correlation is stable across regimes ($\widehat{\xi}_{\text{prof}} \approx$ 0.29–0.31), consistent with the value used in the sampler.

Table 1: Parameter estimation under different sampling methods

| | Sampling method | | |
|---|---|---|---|
| | At random | Stratified by traffic-bins | |
| | | rush | Non-rush |
| $\widehat{\mu}$ | 16.46 | 17.34 | 13.70 |
| $\widehat{\sigma}_{\text{prof}}$ | 25.18 | 26.91 | 18.44 |
| $\widehat{\xi}_{\text{prof}}$ | 0.30 | 0.31 | 0.29 |
| $\widehat{\nu}$ | 1.74 | 2.04 | 1.15 |

A note on $\widehat{\sigma}_{\text{prof}}$. The profile estimator equals $\{\text{Var}_j[\bar{\tau}_j]/\mathbb{E}_j[1/n_j]\}^{1/2}$, where $\bar{\tau}_j$ is the mean edge duration of trip $j$ and $n_j$ its edge count; it charges every between-trip difference in mean edge duration to within-trip edge noise and then rescales by $1/\mathbb{E}_j[1/n_j] \approx 49$. Edge lengths in this network span three orders of magnitude—a median of 109 m against a maximum of 46.6 km—so trips are far from exchangeable, and the 827 edges longer than 5 km, 0.05% of the data, carry 21% of the variance of edge duration. The estimate is therefore governed by how many long links a given subsample happens to contain, and ranges from 48.7 to 375.4 over twenty draws of 500 trips. The pricing sections instead use the $\sigma$ under which predicted and realized travel times have matching spread, 56–61 s on held-out trips, which is deterministic and reproduces the premiums of Table 2. The trip-specific estimates are unaffected: conditioning on the route gives each link its own mean and variance, and the resulting premium moves by less than 1% across subsamples.

## 7.5 Sensitivity to the strike price $K$

We price the on-demand guarantee ($\text{P}_1$) for $10,000$ trips with $r = 0$ and $\zeta = 0$, so that (4.6) reduces to (7.2), and sweep the strike from the expected price ($\times 1$) to a 5% and 10% markup ($\times e^{0.05}$, $\times e^{0.1}$). Table 2 reports the outcome.

$$R(t_0, t_0, \rho, K) = (d_0 - K)\{1 - \Phi(d_1)\} + C_1 d_1 \phi(d_1), \tag{7.2}$$

where

$$d_0 = C_0 + C_1 \mathbb{E}[\mathcal{T}_\rho] + C_2 \delta_\rho, \qquad d_1 = \frac{Ke^{\zeta} - d_0}{C_1\{\text{Var}[\mathcal{T}_\rho]\}^{1/2}}.$$

Raising the strike makes the guarantee less likely to trigger, so the win rate climbs monotonically, from 0.75 to 0.92 for the population model and from 0.67 to 0.95 for the trip-specific model. The average premium charged to the rider falls correspondingly. The premium is

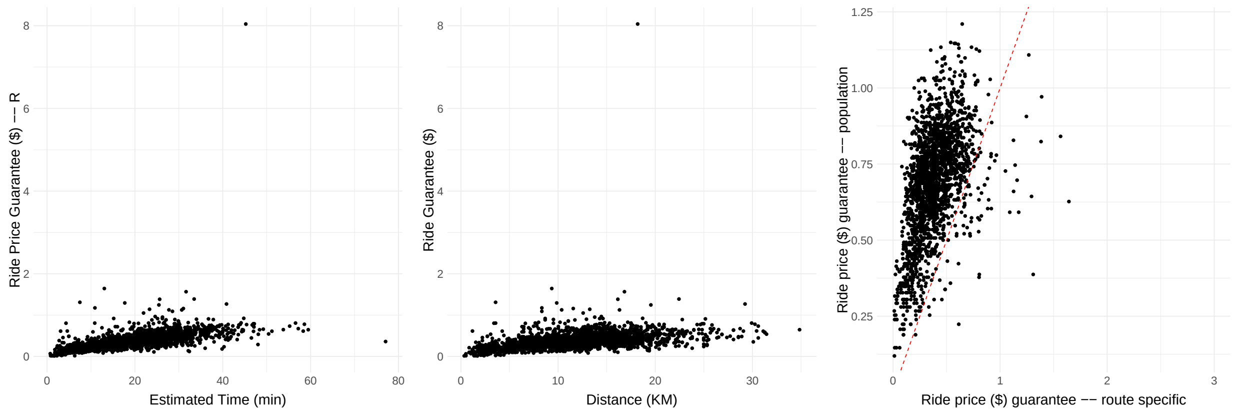


Figure 7: Premiums of on-demand (P$_1$) scenario for 2000 test trips in QCD, while setting $K = P_\rho(s_0, t_0)$, as the expected price at the start (location, time) = $(s_0, t_0)$.

modest throughout, between 0.4% and 4.4% of the trip price. At every strike the trip-specific premium is roughly half the population premium (for instance 2.15% vs. 4.43% at $\times 1$), a direct consequence of the tighter variance obtained by conditioning on the route. Both models are profitable on average (profit factor above one at all strikes), but the population model carries the larger tail risk, with a worst-case single-ride loss near $29 against $22 for the trip-specific model.

Table 2: On-demand price-guarantee performance under different strike prices, with $r = 0$ and $\zeta = 0$, over 10,000 trips. Profit and premium in CAD; profit factor is the ratio of total gains to total losses.

| Model | Strike | % profitable | Avg. profit ($) | Max. loss ($) | Profit factor | Premium/$P$ (%) |
|---|---|---|---|---|---|---|
| Population | $\times 1$ | 0.75 | 0.18 | −28.79 | 1.39 | 4.43 |
| | $\times e^{0.05}$ | 0.84 | 0.11 | −27.58 | 1.41 | 2.35 |
| | $\times e^{0.1}$ | 0.92 | 0.04 | −25.88 | 1.36 | 1.11 |
| Trip-specific | $\times 1$ | 0.67 | 0.04 | −21.85 | 1.12 | 2.15 |
| | $\times e^{0.05}$ | 0.87 | 0.03 | −20.73 | 1.23 | 0.73 |
| | $\times e^{0.1}$ | 0.95 | 0.05 | −19.36 | 1.99 | 0.36 |

These magnitudes translate into very inexpensive guarantees in absolute terms. Consistent with Figure 7, the premium grows essentially linearly in distance and travel time, and for a typical 4 km ride—priced at $K = 3.17 + 0.31 \times (0.0982 \times 4000/60) + 0.9 \times 4 \approx 8.8$ CAD—the trip-specific guarantee costs on the order of 0.2–0.5 CAD. Knowing the realized route therefore prices the guarantee at roughly half the cost of the population approach, with materially smaller downside. The profit profile in Figure 8 matches this picture: maximum profit $R$ is retained whenever the realized price stays below $K$, profit then declines linearly to the break-even point $K + R$, and the loss beyond that point is bounded because real network trips have finite duration. The left panel traces that profile out of sample. Roughly two thirds of the held-out trips settle at the full premium, the realized fare stays within about ±20% of the guarantee, and the worst ride of the 2,000 loses $7.95, so the tail is short in both directions.

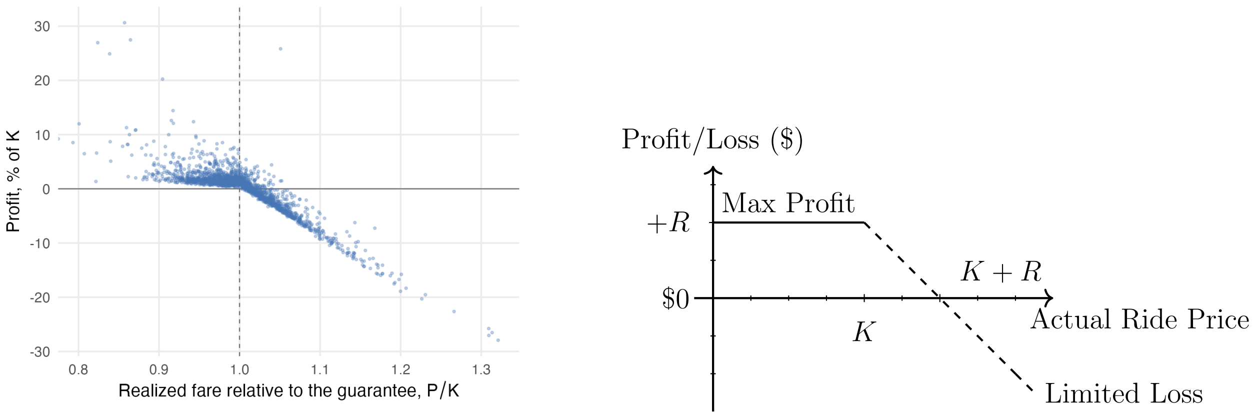


Figure 8: Realized and theoretical profit of the on-demand guarantee. (Left) Per-ride profit against the realized fare, each scaled by that trip's own guarantee $K$, over the 2,000 held-out trips priced with the trip-specific model at $K = P_\rho(s_0, t_0)$, $r = 0$ and $\zeta = 0$; realized fares use the observed edge durations, so nothing on this panel is simulated except the premium. (Right) The theoretical profile: the premium $R$ is retained whenever the realized fare stays below $K$, profit then declines linearly and breaks even at $K + R$.

## 7.6 Discount memberships

We next bundle a flat 10% discount into the guarantee, fixing the strike at a constant fraction of the expected price,

$$K = 0.9P_\rho(s_0, t_1) \tag{7.3}$$

in the guarantee (4.4), and vary the cutoff threshold $\zeta \in \{0, 0.1\}$ separately. We price a 40-ride monthly membership over 100 counterfactual riders. No ride is known at subscription, so we mimic the distribution of potential rides: for each day $y = 1, \ldots, 30$ we draw the number of rides $M_y$, and for each ride a traffic bin $b_k$ and a route $\rho_k$, as

$$\begin{aligned} M_y &\sim \mathsf{Binomial}(2, 2/3), \\ b_k &\sim \mathsf{Uniform}\left[\mathsf{AM\text{-}rush,\ PM\text{-}rush,\ Otherwise}\right], \quad k = 1, \ldots, M_y, \\ \rho_k &\sim \mathsf{Uniform\ over\ rides\ in\ bin}\ b_k\ \mathsf{in\ the\ training\ set.} \end{aligned} \tag{7.4}$$

The Binomial mean leads to $30 \times 2 \times \frac{2}{3} = 40$ expected rides per month; the membership premium is the sum of the per-ride premiums averaged over this distribution.

In every membership table the quantity being scaled is the same—the profit of the guarantee itself, (7.1) summed over the rider's month, with the fares passed through—and only the scale differs. This section and Section 7.7 divide it by the rider's *mean fare*, so that $-16.7\%$ is a loss of one sixth of one typical ride; Section 7.8 divides it by the month's *total* fares instead. The two differ by the ride count, so a figure quoted per average fare is the corresponding share of spend multiplied by the number of rides. Table 3 shows that the discount reverses the ranking observed in the on-demand case. The population model stays slightly profitable, with an average return of 2.7–2.9%, whereas the thinner trip-specific premium no longer covers the discounted guarantee and turns marginally loss-making, at $-0.4$ to $-0.5\%$. Introducing a non-zero cutoff $\zeta = 0.1$ lowers the premium charged to the rider but also erodes the provider's margin and deepens the worst-case loss, since fewer overruns are recovered. This suggests that the cost-based premium has little slack once a fixed discount is layered on top: the route conditioning that makes the on-demand guarantee cheap also leaves the discounted trip-specific product without a buffer against realized overruns.

Table 3: Discount-membership performance with $K = 0.9\,P$ under different cutoff thresholds $\zeta$, over 100 trips.

| Model | Cutoff $\zeta$ | Profit per avg. fare (%) | Avg. profit (\$) | Max. loss (\$) | Premium/$P$ (%) |
|---|---|---|---|---|---|
| Population | 0 | 2.86 | 0.09 | $-8.41$ | 11.42 |
| | 0.1 | 2.71 | $-0.03$ | $-9.28$ | 9.73 |
| Trip-specific | 0 | $-0.45$ | $-0.23$ | $-6.90$ | 10.49 |
| | 0.1 | $-0.53$ | $-0.36$ | $-7.99$ | 7.90 |

## 7.7 Membership abuse

Riders may change behaviour once a flat-price membership is held, favouring longer routes or peak hours. We stress-test the membership by constructing 100 counterfactual riders of 40 trips each, drawn with replacement from the test trips, and forcing a fraction $M_1/40$ of each rider's trips to come from the upper-$q$ quantile of the trip-length distribution. The fraction $M_1/40$ controls how *often* the rider abuses the membership; the quantile $q$ controls how *severe* each abusive trip is ($q = 0.9$ draws only from the longest 10% of trips). For example $M_1 = 10$, and $q = 0.9$ would sample (with replacement) 10 rides (out of 40) from the highest 10% of rides in the test trip. With no abuse ($M_1/40 = 0$, the top row of Table 4), the severity quantile $q$ is irrelevant—no trip is drawn from the tail—so that row is nine *independent* Monte-Carlo estimates of one no-abuse baseline rather than nine distinct quantities. The variability is a result of fresh resample of the rider's 40 trips and of their simulated travel times, and they scatter around a common mean of about 1.7% with standard error of 0.36%. Figure 9 therefore drops that row and reports the no-abuse baseline in its horizontal-axis label instead.

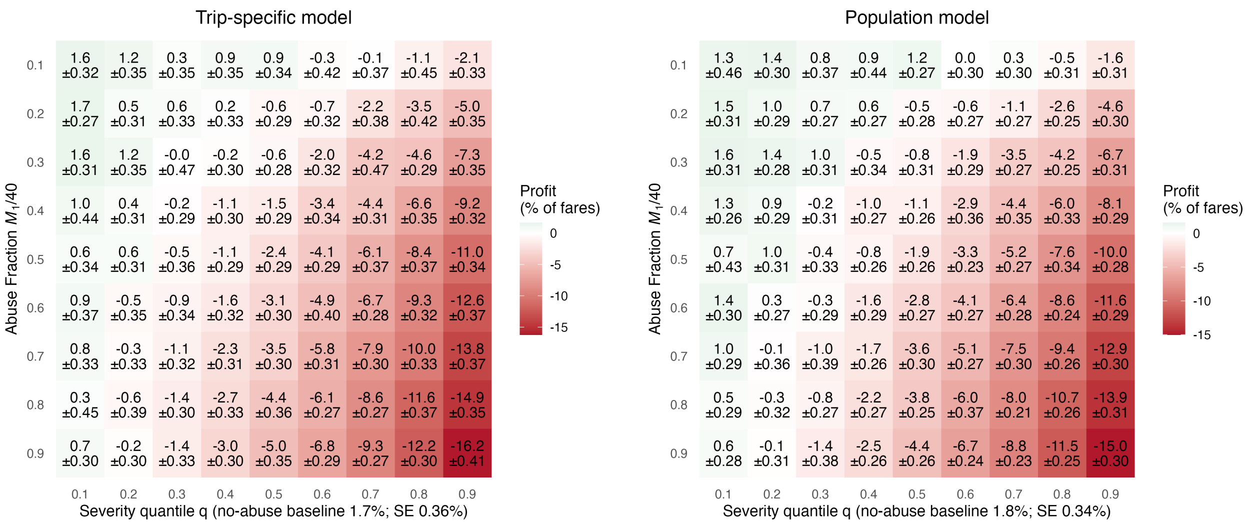


Figure 9: Membership-abuse stress test: average profit per average fare (%) over 100 counterfactual riders of 40 trips

Figure 9 reports the average profit per average fare and its standard error across the abuse grid. Profit declines monotonically in both $M_1/40$ and $q$, and the two factors compound: moderate abuse ($M_1/40 \leq 0.5$, $q \leq 0.5$) keeps the guarantee within about 2.5% of fares of break-even, and even $(0.5, 0.6)$ costs only $-4.1\%$, whereas the joint extreme $(0.9, 0.9)$ drives it to $-16.2\%$ under the trip-specific model ($-15.0\%$ under the population model), mapping the negative tail of the product. The standard errors are stable at 0.3–0.5% across the whole grid, so the ordering of neighbouring cells is resolved well within Monte-Carlo noise. Overall the surface shows graceful degradation: the pricing remains resilient under mild-to-moderate abuse and only collapses when a large share of trips are simultaneously drawn from the extreme tail, which gives the provider an explicit, tunable handle on risk tolerance.

Table 4: Standard error (%) of the average profit per average fare in Figure 9 under the trip-specific model, over the 100 counterfactual riders. The $M_1/40 = 0$ row is the no-abuse baseline, which Figure 9 reports in its axis label rather than as a row.

| $M_1/40$ | $q$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
| 0 | 0.42 | 0.35 | 0.35 | 0.31 | 0.44 | 0.36 | 0.41 | 0.32 | 0.30 |
| 0.1 | 0.32 | 0.35 | 0.35 | 0.35 | 0.34 | 0.42 | 0.37 | 0.45 | 0.33 |
| 0.2 | 0.27 | 0.31 | 0.33 | 0.33 | 0.29 | 0.32 | 0.38 | 0.42 | 0.35 |
| 0.3 | 0.31 | 0.35 | 0.47 | 0.30 | 0.28 | 0.32 | 0.47 | 0.29 | 0.35 |
| 0.4 | 0.44 | 0.31 | 0.29 | 0.30 | 0.29 | 0.34 | 0.31 | 0.35 | 0.32 |
| 0.5 | 0.34 | 0.31 | 0.36 | 0.29 | 0.29 | 0.29 | 0.37 | 0.37 | 0.34 |
| 0.6 | 0.37 | 0.35 | 0.34 | 0.32 | 0.30 | 0.40 | 0.28 | 0.32 | 0.37 |
| 0.7 | 0.33 | 0.33 | 0.32 | 0.31 | 0.30 | 0.31 | 0.30 | 0.33 | 0.37 |
| 0.8 | 0.45 | 0.39 | 0.30 | 0.33 | 0.36 | 0.27 | 0.27 | 0.37 | 0.35 |
| 0.9 | 0.30 | 0.30 | 0.33 | 0.30 | 0.35 | 0.29 | 0.27 | 0.30 | 0.41 |

## 7.8 Commuter memberships

Consider a monthly membership ($P_3$) of $M = 30$ rides between a fixed origin and destination, departing at a fixed hour of the day, with the price of *each* ride capped at a promised $K$ that is quoted once and held for the whole month. Origin, destination, and departure time are known at subscription. Traffic and route uncertainity remain unknown at time of subscription.

We construct a counterfactual commuter from an origin–destination pair of QCD trips holding at least $h + 1$ trips, where $h$ is the number of trips that are already observed. One trip on the pair is *held out* and the remaining $h$ form the commuter's *history*. A departure time is drawn from the pair's trips and every trip on both sides is shifted to it, so that the whole month is priced and realized at the same hour and traffic bin. The month's $M$ rides are $M$ route randomizations (5.2) of the held-out trip, each carrying an independently sampled travel time (5.3); the membership itself is priced from (4.8) with $r = 0$, with $\pi(t_0)$ degenerate at the commuter's departure time. Route sampling is applies to the history $h$ 10 times, where the strike and premium is averaged across those samples. Since the member is promised one cap for the month:

$$\mathsf{R}_{\mathsf{mem}}(t, M) = M\,\bar{R}\,(t_0, t_0, K)\,, \qquad \bar{R}\,(t_0, t_0, K) = \frac{1}{|\Pi_h|} \sum_{\rho_i \in \Pi_h} R\,(t_0, t_0, \rho_i, K)\,, \tag{7.5}$$

where $\Pi_h$ collects the sampled history routes. The member pays $\mathsf{R}_{\mathsf{mem}}(t, M)$ up front and then $\min(P_i, K)$ for each ride $i$ actually taken, so the cap binds ride by ride and the provider absorbs the overrun $(P_i - K)^+$; there is no separate cap on the month's total. Profit is (7.1) accumulated over taken rides (30 at max), at cutoff $\zeta = 0$, against the single monthly premium. Held-out trips are excluded from the travel-time model fit, so no membership is priced by a model that has seen the rides it will be judged against.

We draw 100 such commuters for each history size $h \in \{2, 4, 6, 8\}$, and price under both travel-time models at the plain cap $K = P$ and at the discounted cap $K = 0.9P$ of Section 7.6. Table 5 reports the outcome.

Table 5: Commuter-membership performance over 100 counterfactual commuters holding a 30-ride monthly membership on a fixed origin–destination pair at a fixed departure hour, where $h$ is the number of past trips on the pair used to price the membership.

| Model | Strike | $h$ | Profit (% of fares) | SE | Avg. profit ($) | Max. loss ($) | Win rate |
|---|---|---|---|---|---|---|---|
| Population | $P$ | 2 | 2.15 | 0.46 | 10.86 | −86.8 | 0.77 |
| | | 4 | 1.71 | 0.60 | 9.62 | −199.5 | 0.78 |
| | | 6 | 2.07 | 0.53 | 10.57 | −157.2 | 0.83 |
| | | 8 | 2.13 | 0.63 | 10.91 | −238.7 | 0.79 |
| | $0.9\,P$ | 2 | 3.57 | 0.73 | 18.92 | −110.5 | 0.71 |
| | | 4 | 2.79 | 0.83 | 15.34 | −213.7 | 0.68 |
| | | 6 | 3.09 | 0.76 | 16.17 | −177.7 | 0.73 |
| | | 8 | 3.94 | 0.88 | 22.85 | −252.1 | 0.75 |
| Trip-specific | $P$ | 2 | 0.00 | 0.40 | −0.54 | −130.7 | 0.65 |
| | | 4 | −0.12 | 0.50 | 1.84 | −193.7 | 0.67 |
| | | 6 | −0.39 | 0.50 | −3.32 | −210.5 | 0.66 |
| | | 8 | −0.31 | 0.59 | −1.96 | −258.3 | 0.64 |
| | $0.9\,P$ | 2 | 0.23 | 0.57 | 0.78 | −155.2 | 0.56 |
| | | 4 | −0.17 | 0.63 | 2.36 | −205.4 | 0.51 |
| | | 6 | −0.55 | 0.60 | −4.37 | −225.1 | 0.55 |
| | | 8 | −0.08 | 0.74 | 0.42 | −278.3 | 0.55 |

Membership seems fairly priced under route conditioning and carries a small margin without it. At a low monthly cost, all memberships can be profitable. The trip-specific product clears between −0.6% and +0.2% of the member's fares across the whole grid. All are within one standard error from zero, on a premium of 3.7% of those fares at $K = P$ and 10.8% at $K = 0.9P$. The population product clears +1.7% to +3.9% on a premium of 5.4% and 12.1% respectively.

The population model has a larger variance, which adds a buffer to profit margins. The discounted strike is not markedly worse than the plain cap. The premium absorbs the discount almost exactly, with the mean realized ride price landing between 0.995 and 1.008 times the strike at $K = P$ and between 1.105 and 1.121 at $K = 0.9P$, against the $1/0.9 = 1.111$ that the discounted cap implies.

Figure 10 shows that the margin depends on the number of rides taken. The distribution of profit is tight around zero in both models, and the difference in mean profit is carried by a left tail of a handful of members per cell—about 2% of them—whose overruns above the cap $K$, accumulated over the month, exceed the premium collected by 20% to 40% of their fares. The typical loss-making member is far milder: overruns of some 12% of fares against a premium of 10%, for a net loss of 2 to 4%. The guarantee is therefore doing what it is priced to do on the typical commuter and is decided, in aggregate, by a small number of bad months.

The history size leaves no systematic trace. The premium ratio is flat in $h$ (3.5–3.7% for the trip-specific model, 4.8–5.0% for the population model) and the returns do not trend, so what moves between rows is which origin–destination pairs the narrowing eligibility admits rather than the amount of information priced on. The worst-case loss does grow with $h$, from about \$100–130 to \$245–280 per member, but for the same reason: the pairs that carry eight or more trips are the busier, longer commutes.

A member pays up front and need not travel every day, and because the premium is collected regardless, unused rides are pure breakage (Figure 11). Averaged over the grid at $K = P$, the trip-specific return rises from −0.2% at full use to +0.7% when four rides in five

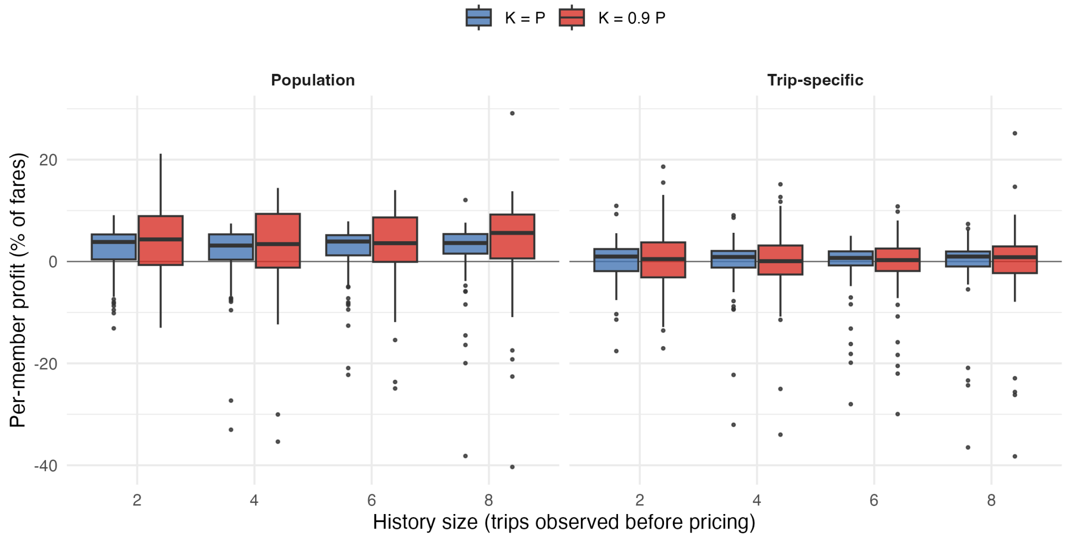


Figure 10: Per-member profit on the commuter membership, as a percentage of the member's fares, over 100 counterfactual commuters, by travel-time model, strike, and history size $h$. The population model sits above zero throughout while the trip-specific model sits lower; both carry a thin left tail of members whose realized month runs well past the promised cap.

are taken and +2.2% at three in five, and the population return from +1.0% to +4.3%; win rates rise from 0.65 to 0.84 and from 0.72 to 0.84. The provider's exposure is therefore worst against the fully utilizing member, which is the case reported in Table 5, and any realistic attrition moves the product into profit. Breakage is also the one channel that works in the provider's favour under the discounted cap: at $K = 0.9P$ the same drop in utilization is worth two to three times as much, because the premium being kept is itself two to three times larger.

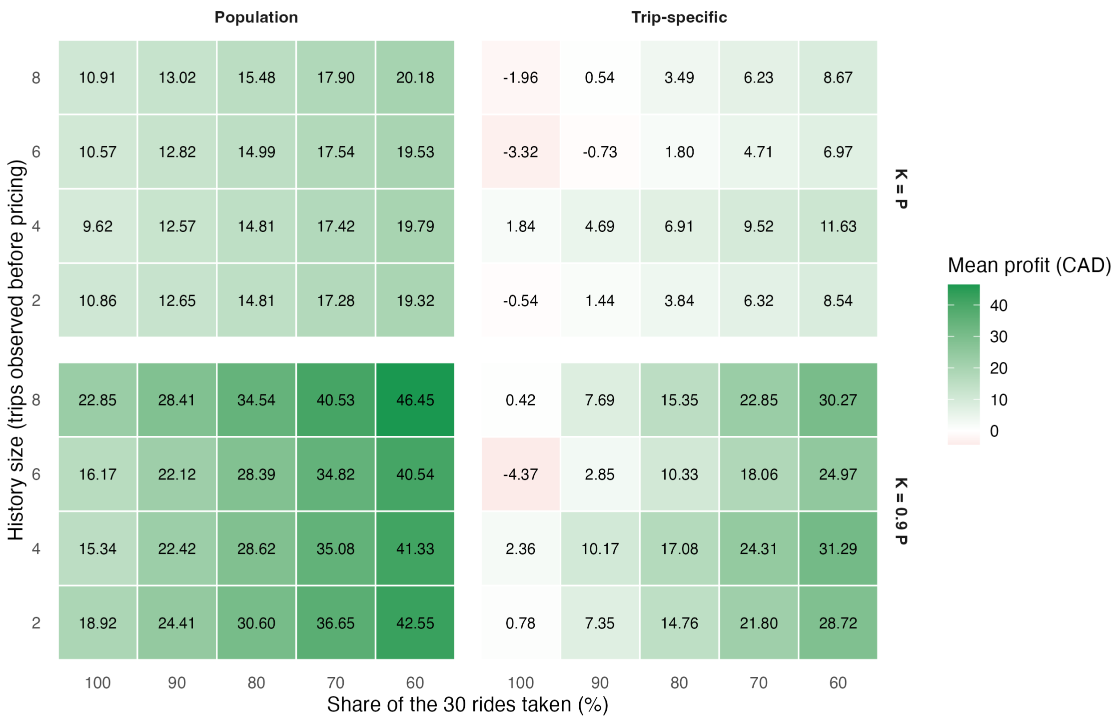


Figure 11: Breakage in the commuter membership: mean profit per member (CAD, over a 30-ride month) as the share of committed rides actually taken falls from 100% to 60%. The premium is collected in full regardless, so unused rides accrue directly to the provider.

The commuter product is also where QCD is thinnest. No origin–destination pair in the data carries more than 13 trips, and the number of pairs able to supply a commuter falls from 1,001 at $h = 2$ to 63 at $h = 8$, so the larger history sizes draw repeatedly from a small set of pairs. The history axis can be read as a direction, not as an extrapolation to the data-rich commuter a large platform would actually observe.

# 8 Discussion

Providers of on-demand transportation build increasingly complex models to find a sweet spot between profitability and competitiveness. These models typically rely on point estimates of travel time and leave travel-time uncertainty out of the price. Integrating that uncertainty has proven elusive for two reasons: analytical pricing tools for it have been lacking, and the travel-time distribution itself is difficult to characterize on a real network. This work supplies both pieces. Taking the asymptotic travel-time law of Elmasri et al. [2023] as an arithmetic Brownian motion on a metric graph, we priced the right to a capped fare as a contingent claim, obtaining a closed-form, Black–Scholes-type premium (4.6) under both a population and a route-conditional travel-time model, and we packaged that single premium into on-demand, scheduled, and membership products.

Empirically, the guarantee is inexpensive. On QCD the premium runs between 0.4% and 4.4% of the fare across strikes, and conditioning on the realized route roughly halves it—for example 2.15% versus 4.43% at the expected-price strike—while also shrinking the worst-case single-ride loss (near \$22 versus \$29). The route-conditional model is therefore the better product for the on-demand guarantee: tighter variance yields a cheaper premium with a thinner tail. That same tightness reverses once a flat discount is layered on: the discounted trip-specific membership loses its buffer against overruns and turns marginally loss-making, while the coarser population model retains a small margin. The lesson is that the route conditioning which makes the bare guarantee cheap also removes the slack a bundled discount needs, so the two design choices interact and should not be tuned in isolation.

A unifying way to read the membership results is that abuse exploits precisely the dimension along which the price fails to condition. The unrestricted flat membership conditions on nothing: a single strike $K$ covers any route at any time, so the rider's marginal price of distance is zero and a rational holder substitutes toward the longest covered trips, with the provider's per-ride loss approaching $d_0 - K$, the full price gap on a long ride. The commuter membership fixes the origin–destination pair and the departure hour, and thereby conditions on both length and timing; nothing but travel-time variance on a known commute is left for the holder to select on. Priced that way it is close to fair rather than exposed: over the grid of Table 5 the route-conditional product returns between $-0.6\%$ and $+0.2\%$ on a premium of 3.7% of the member's fares, and any shortfall in utilization moves it into profit. We have not stress-tested it against a holder who deliberately abuses the remaining channel, so we claim the absence of a behavioural channel by construction, not a measured reduction in exposure. Per-ride pricing conditions on everything, leaving no such channel at all: each ride is an independently priced guarantee, and selection cannot generate expected losses. Read together, the abuse grid and these three products trace a single axis—the more the price conditions, the less room a rational holder has to select against it.

This logic suggests a class of products that are abuse-proof by construction: those whose subsidy is additive in the fare rather than attached to its variable part. Consider a bundle granting a fixed \$$D$ discount on each of $N$ rides. Because the discount is a constant, it leaves the rider's marginal price of distance and time unchanged; there is no incentive to lengthen or retime trips. The provider's liability is then deterministic and bounded,

$$\text{Cost} \;=\; \sum_{i=1}^{N} \min(D, P_i) \;=\; ND - \sum_{i=1}^{N} (D - P_i)^+ \;\leq\; ND, \tag{8.1}$$

with expected value $\mathbb{E}[\text{Cost}] = ND - N\,\mathbb{E}[(D-P)^+]$. The only stochastic term is the correction for rides whose fare falls below $D$, on which the non-negativity of price caps the rebate at the fare itself. No travel-time model is required to price the product—only the fraction of fares below $D$—and the maximum liability is known in advance. Moving the subsidy from the level of the fare (a capped total, marginal length-price zero) to an additive per-ride amount (marginal length-price preserved) converts an uninsurable moral-hazard problem into a bounded, closed-form cost. Product design, not pricing sophistication, is what neutralises the abuse.

These conclusions inherit the assumptions under which they were derived, and their limits mark the boundary of the claims. Our pricing is deliberately cost-based and set in a monopoly market with immediate fulfilment (A1–A3): we price the guarantee written on top of the fare process rather than the market-clearing fare itself, and we exclude the matching, strategic-driver, and supply–demand-equilibrium mechanisms that a two-sided model would carry. Riders are taken to be price-insensitive beyond the modelled price (A4), so the abuse analysis captures selection on route and timing but not elastic demand response; a rider who takes more or fewer trips because the membership exists is outside the present model. The frictionless assumptions (B1–B2)—constant known risk-free rate, no fees or taxes—make the discounting exact but abstract from real settlement costs. Finally, because road travel time is not a traded, hedgeable asset, the market is incomplete and no replicating portfolio exists; our premium is a discounted expected cost under the modelled law, not a no-arbitrage replication price, and it is therefore only as good as that law's fit—which the out-of-sample validation of Section 7.2 supports on QCD but does not guarantee on networks with different traffic structure.

Several extensions follow naturally. Relaxing price-insensitivity (A4) would let the framework speak to demand response and to the joint design of price and membership, at the cost of the two-sided machinery we set aside. Incorporating a non-zero and stochastic interest rate (B1) would matter for genuinely long-dated products—scheduled rides booked months ahead, or memberships priced a year out—where the value-of-money adjustment stops being negligible. On the data side, the samplers of Section 5 make the method portable to any network with edge-level travel-time statistics, so replicating the QCD study on a second city would test how far the premium magnitudes and the abuse geometry generalize. Most promising, the additive-subsidy result points toward a broader design question: given a target subsidy and a tolerance for selection, which contract shapes are both attractive to riders and bounded for the provider? The guarantee, the capped membership, and the fixed per-ride discount are three points on that spectrum; characterizing it fully is left to future work.

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